

2 (a) 1.6 ± 0.2 cm B1 [1]

(b) $1.6 / 50 = 0.032$...(ignore any uncertainties)..... B1 [1]

(c) idea of adding fractional uncertainties C1

$$(0.2 / 1.6) + (0.1 / 50)$$

$= 0.127$ OR 12.7% ...(-2 marks if uncertainties not added) A1

actual uncertainty $= (\pm) 0.004$ A1 [3]

(do not allow more than 2 sig. fig)

2 (c) $x^2 = y^2 + z^2$ OR use of triangle etc C1